S2: Advanced Statistical Methods for DataScience – Coursework

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1 Introduction

This report is submitted as part of my solutions to the assignment coursework. I used autogeneration tools such as Chat-GPT to learn how to calculate the inferred rate for this specific case and to compare my implementation of the height_change _move() function in RJMCMC.py. Then my thinking was based on these findings to insert my own implementation based on derivations from [3] as asked in the exercise.

2 Answers

Exercise 1: Accident Data.

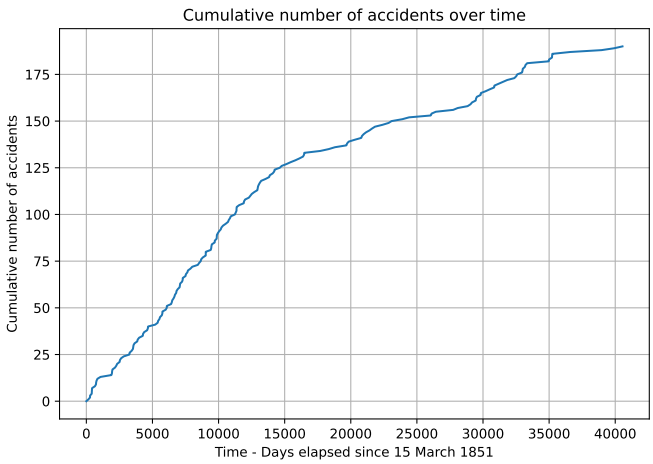

Figure 1: Cumulative number of accidents for the released coal mining acidents data

The mean rate of accidents over the total period covered is given by:

$$\frac{\mathrm{N}}{\mathrm{L}} = \frac{191}{40550} = 0.004710 \text{ accidents/day}.$$

Exercise 2: Priors.

(a) The difference between plain order statistics prior and even-numbered order statistics prior is that the former places change points uniformly across the interval $[0, \mathrm{L}]$ with no preference for spacing allowing change points to possibly fall very close together (clustering) as it creates less

regular spacing between them. This may lead the model to create very small gaps between change points.

On the other side, even-numbered order statistics avoids clustering by producing reasonable spacing between changepoints ensuring that one extra random point lies between two change point positions making them more evenly spread out.

As it can be seen in Figure 2 of smallest gaps from both order statistics priors, the distribution of smallest gaps for plain order statistics prior has its peak very close to zero meaning that the density of points very close to each other is very high compared to that of even-numbered order statistics prior where the distribution of smallest gaps is more shifted to the right pushing the change point positions apart and preventing from clustering and from having intervals that are too small to be statistically significant.

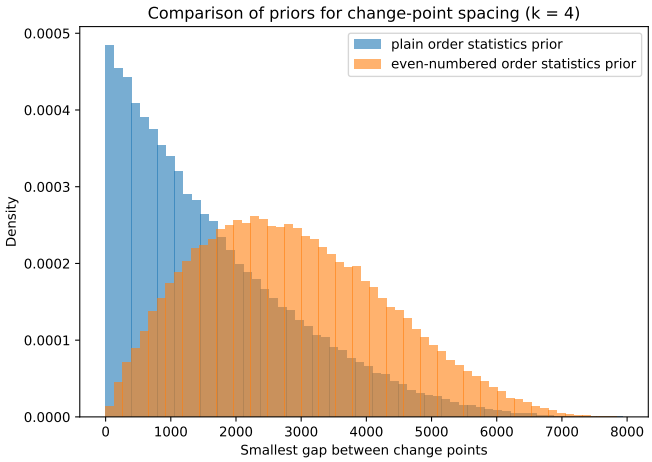

Figure 2: Small gaps distributions for plain order statistics and even-numbered order statistics priors

(b) In our setup, we start by drawing u from a uniform distribution on $[0, L]$ independently:

$$u_1, u_2, \ldots, u_{2k+1} \sim \text{Uniform(0,L)}$$

so each random variable u_i has density

$$f(u_i) = \frac{1}{L}, \ 0 < u_i < L.$$

The joint PDF of all $2k+1$ plain order statistics is:

$$\begin{aligned} f(u_1, u_2, \ldots, u_{2k+1}) &= \prod_{i=1}^{2k+1} \frac{1}{L} \\ &= \frac{1}{L^{2k+1}} \end{aligned} \tag{1}$$

Given that the PDF of the plain order statistics is constant, on the region $0 < u_{(1)} < u_{(2)} < \cdots < u_{(2k+1)} < 1$, given by

$$P\left(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}\right) = (2k+1)!$$

on the region $0 < u_{(1)} < u_{(2)} < \cdots < u_{(2k+1)} < L$, the PDF $P_1\left(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}\right)$ of the plain order statistics is given by

$$\begin{aligned} P_1\left(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}\right) &= P\left(u_{(1)}, u_{(2)}, \ldots, u_{(2k+1)}\right) \times f(u_1, u_2, \ldots, u_{2k+1}) \\ &= (2k+1)! \times \frac{1}{L^{2k+1}} \\ &= \frac{(2k+1)!}{L^{2k+1}} \end{aligned}$$

Using our notation for change points, ie $u_{2j} = s_j$, the above PDF can be rerwritten as:

$$P_1\left(u_{(1)}, s_1, u_{(3)}, s_2, \ldots, s_k, u_{(2k+1)}\right) = \frac{(2k+1)!}{L^{2k+1}}$$

The even-numbered order statistics prior PDF is then the marginal PDF over the odd-numbered order statistics. This marginal PDF is:

$$\begin{aligned}
\pi(s_1, \ldots, s_k | M_k) &= \int P_1\left(u_{(1)}, s_1, u_{(3)}, s_2, \ldots, s_k, u_{(2k+1)}\right) du_1 du_3 \ldots du_{2k+1} \\
&= \frac{(2k+1)!}{L^{2k+1}} \int_0^{s_1} du_1 \int_{s_1}^{s_2} du_3 \cdots \int_{s_k}^{L} du_{2k+1} \\
&= \frac{(2k+1)!}{L^{2k+1}}\, s_1 \cdot (s_2 - s_1) \cdot \cdots \cdot (L - s_k) \\
&= \frac{(2k+1)!}{L^{2k+1}}\, g_1 \cdot g_2 \cdot \ \cdots \ \cdot g_{k+1} \\
&= \frac{(2k+1)!}{L^{k+1}} \prod_{j=1}^{2k+1} g_j \\
&= \frac{1}{L^{k+1} \dfrac{1}{(2k+1)!}} \prod_{j=1}^{2k+1} g_j. \\
&= \frac{1}{L^{k+1} B(\alpha)} \prod_{j=1}^{2k+1} g_j.
\end{aligned}$$

With $\mathrm{B}(\alpha) = \dfrac{1}{(2k+1)!}$ the multivariate beta function evaluated at $\alpha = (2, 2, \ldots, 2) \in \mathbb{R}^{k+1}$. Hence,

$$\pi(s_1, \ldots, s_k | M_k) \propto \prod_{j=1}^{k+1} g_j.$$

Exercise 3: The Constant Rate Model, M_0

(a)

(b) The log evidence for M_0 is given by :

$$\log Z_0 = -1261.0957190056931$$

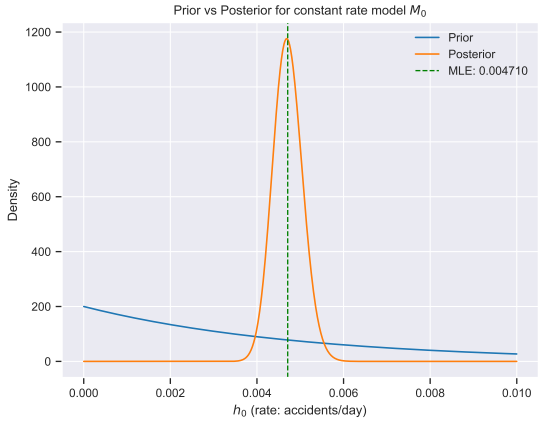

Figure 3: Comparison between the prior and the posterior of the constant rate model M_0, with indication of MLE.

Exercise 4: The 1 Change Point Model, M_1.

(a) For this 1 changepoint model, I used the emcee Python package, which implements the Affine-Invariant Ensemble Sampler (AIES) [1]. This algorithm is particularly robust for highly correlated parameter spaces and multi-modal posteriors, as it leverages an ensemble of walkers to determine proposal distributions dynamically, eliminating the need for manual step-size tuning [2].

Algorithm Settings

The sampler was initialized with 32 walkers to ensure sufficient exploration of the density. I ran the chain for $10,000$ steps per walker, discarding the first $2,000 steps$ as burn-in to remove bias from initial conditions. This yielded a total production chain of $256,000$ samples after burn-in.

Convergence Diagnostics

To ensure the validity of the posterior estimates, the following diagnostics were performed:

- Acceptance Fraction : The mean acceptance fraction was 0.525 While the theoretical ideal for AIES is approximately $0.20 - 0.50$ [**Gelman1**], a value of 0.525 indicates a healthy, efficient exploration without excessive "stickiness."
- Integrated Autocorrelation Time (τ): Calculated as $\tau \approx (46, 51, 46)$ for (s_1, h_0, h_1). The total chain length exceeds 50τ, a common threshold for asserting that the chain has "forgotten" its starting position and converged to the target distribution [4].
- Effective Sample Size (ESS): By thinning the chain according to τ, I achieved an ESS of 8,000 per walker. This provides a sufficiently low standard error for the posterior mean estimates.

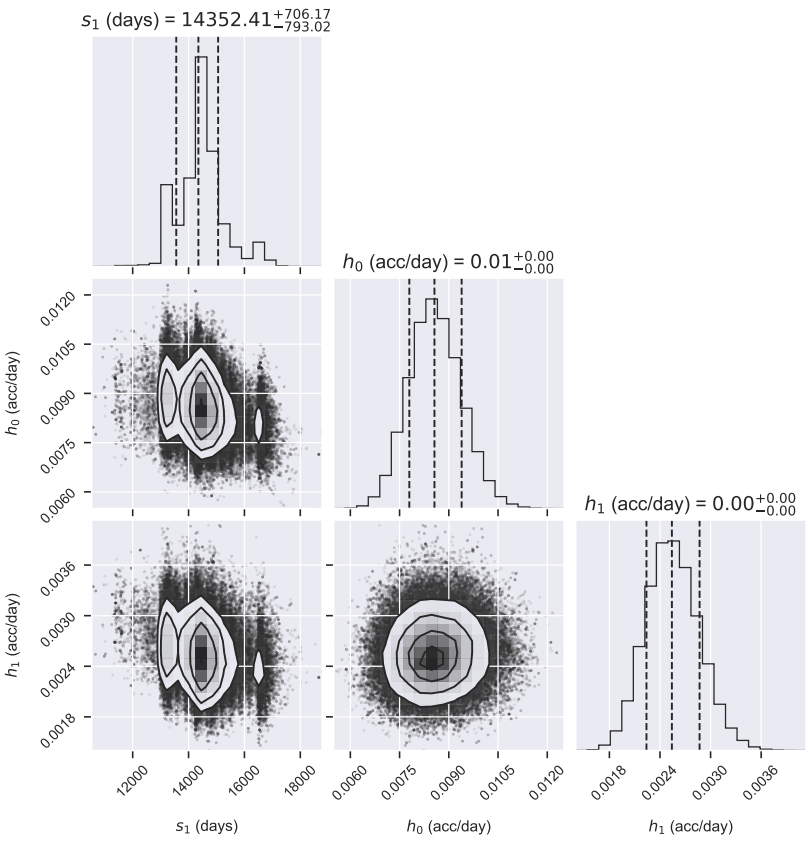

Figure 4: Corner plots for the 1 change model posterior.

(b) The Savage Dickey density ratio allows calculation of the Bayes factor between nested models. We can sse that the constant rate model M_0 can be recovered from M_1 by setting $h_1 = h_0$, hence the change point s_1 becoming irrelevant.

Under this hypothesis, the Savage Dickey density ratio is then given by

$$\mathrm{B}_{0,1} = \frac{Z_{M_1}}{Z_{M_0}} = \frac{P(h_1 = h_0|\{I_i\}, M_1)}{P(h_1 = h_0|M_1)}$$

Difficulty for this problem

The constant rate model M_0 is recovered from M_1 when $h_0 = h_1$, not at a specific point but along an entire surface in the 3D parameter space (s1, h0, h1). The position s_1 being irrelevant when $h_0 = h_1$, we have a 1-dimensional manifold where M_0 is embedded in M_1.

To use the Savage Dickey ratio, we would need to evaluate the prior and posterior densities along this surface $\{(s_1, h, h) : s_1 \in [0, L], h > 0\}$ and integrate over it, which is more complex than the standard point-wise application. Also, as shown in Figure (5) , the posterior density is near-zero on this surface, making numerical estimation challenging.

(c) As the Laplace approximation estimates the evidence by approximating unormalised posterior $P(h_j, s_j|\{I_i\}, M_1)$ as Gaussian around its maximum using:

$$Z_1 \approx P(\hat{h}_j, \hat{s}_j|\{I_i\}, M_1)\sqrt{\frac{(2\pi)^D}{\det C}}$$

where $(\hat{h}_j, \hat{s}_j)$ is the MAP estimate, here $D = 3$ is the dimensionality, and C is the Hessian at the MAP.

Difficulties

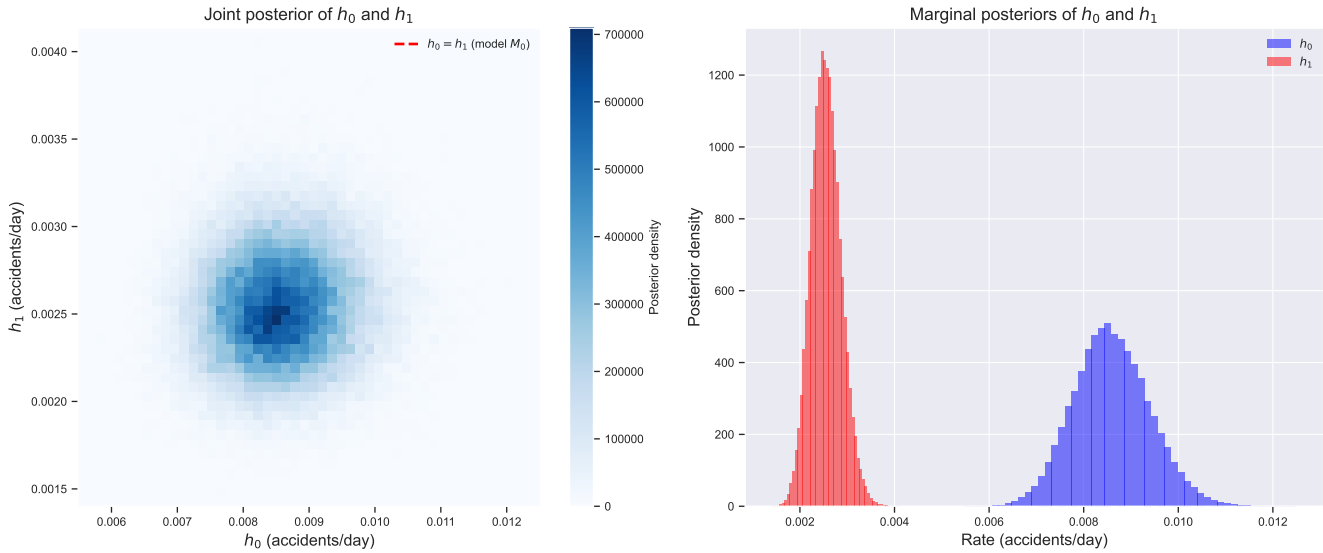

Figure 5: Left: Joint posterior of h_0 vs h_1 showing the constraint surface $h_0 = h_1$ (red dashedline) where M_0 is recovered the posterior having negligible density along this line. Right: Marginal posteriors show clear separation between h_0 and h_1, with $h_0 > h_1$ strongly supported

- Multimodality: The posterior for M_1 can have multiple modes. If there are multiple change points with similar support, also the posterior could have multiple peaks but the Laplace approximation only captures one mode.
- Non-Gaussian shape: The posterior over s_1 has hard boundaries at 0 and L, and maybe skewed or have long tails, violating the Gaussian assumption. The even-numbered orderstatistics prior has a specific functional form that produces non-Gaussian posteriors.
- Prior sensitivity: The even-numbered order statistics prior vanishes at the boundaries, which affects the shape near the edges and may not be well-captured by a Gaussian.

These issues may lead the Laplace approximation to significantly under- or over-estimate the evidence Z_1.

(d) For the 1 change point model M_1, I use the dynesty package, which implements dynamic nested sampling specifically designed for evidence calculation and multi-modal posteriors handling.

Settings

Number of live points	1,000
Bounding method	multi-ellipsoidal
Sampling method	random walk
Termination criterion	$\Delta \log Z < 0.1$

Results

$\log Z_1 = -1187.230775$

With $\log Z_0 = 1261.095719$ from question 3, the Bayes factor is given by

$$\log B_{10} = \log Z_1 - \log Z_2 = -1187.230775 + 1261.0957 = 73.864944 \Rightarrow B_{10} = 1.199880e + 32$$

Posterior Odds Ratio

Assuming equal prior probabilities $P(M_0) = P(M_1) = 0.5$, the posterior odds ratio is:

$$\frac{P(M_1|\{I_i\})}{P(M_0|\{I_i\})} = B_{10} \times \frac{P(M_1)}{P(M_0)} = B_{10} = 9.737867e31$$

which means that model the 1 change point model M_1 is overwhelmingly preferred over the constant rate model M_0. This confirms that there is asignificant change in the accident rate during the observation period.

Exercise 5: The k Change Point Models, M_k.

(a) Completed in RJMCMC.py

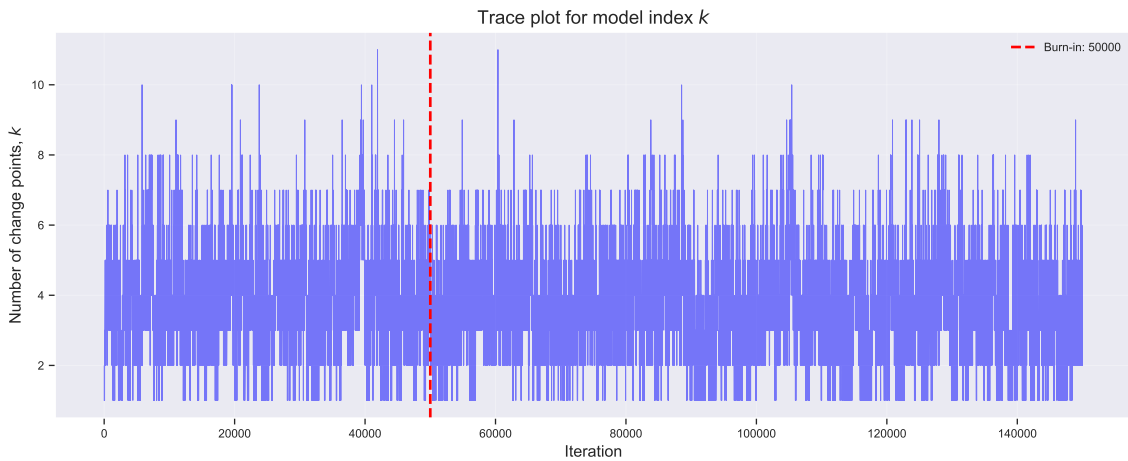

Figure 6: Trace plot for model index k showing how the chain explored the model space

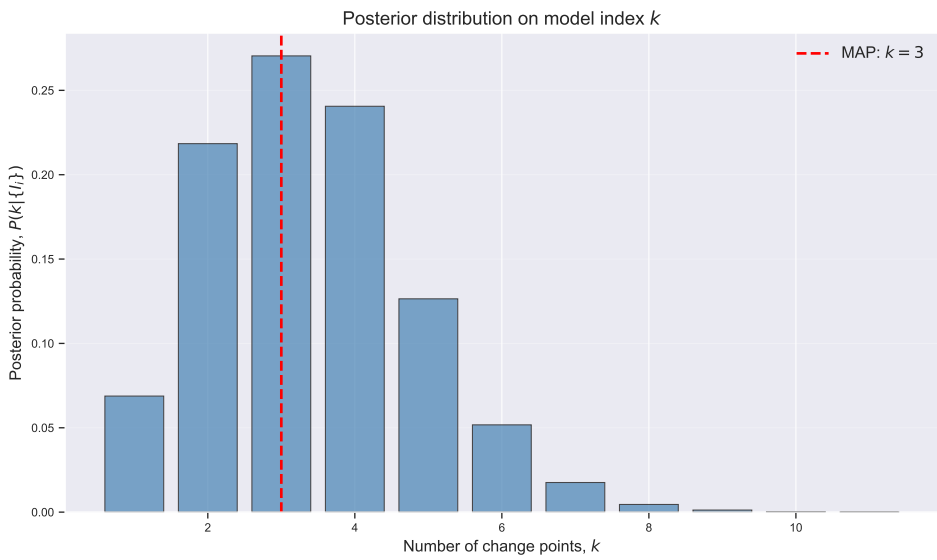

Figure 7: Posterior on the model index k indicating the MAP number of change points (red-dased line)

(b)

The **maximum a posteriori** number of change points is $k = 3$.

(c) The inferred accident rate $x(t)$ (see Figure 8), is calculated by model averaging over all values of k explored by the RJMCMC chain. For each time point t:

I started by collecting the rate $x(t)$ from each MCMC sample (after burn-in). Each sample having a specific value of k, change points $\{s_j\}\}$, and rates $\{h_j\}$. For suc a sample, $x(t)$ is the rate of the segment containing t. Finally, I calculated percentiles acccross all samples at each time t.

The 50% credible interval contains the middle 50% of the posterior mass (25th to 75th-percentiles), while the 90% interval contains 90% of the mass (5th to 95th percentiles). Theseintervals account for both uncertainty within each model Mk and uncertainty about whichmodel is correct (model uncertainty).

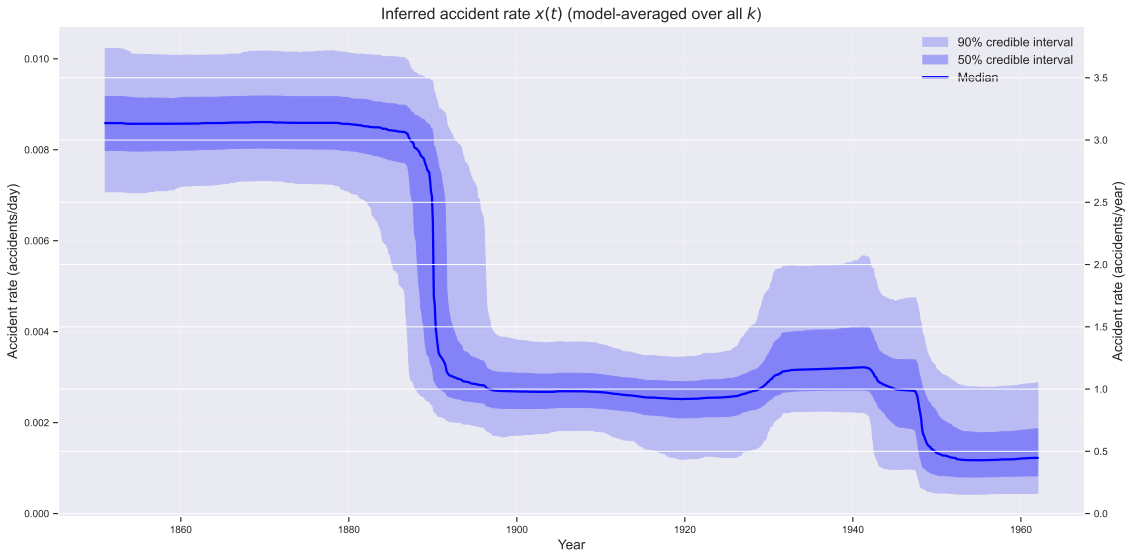

Figure 8: Inferred accident rate $x(t)$ obtained through Bayesian Model Averaging. The blue line denotes the posterior median. The shaded areas represent the 50% and 90 credible intervals.

References

[1] Daniel Foreman-Mackey et al. "emcee: The MCMC Hammer". In: *Publications of the Astronomical Society of the Pacific* 125.925 (Mar. 2013), p. 306. DOI: 10.1086/670067.

[2] Jonathan Goodman and Jonathan Weare. "Ensemble samplers with affine invariance". In: *Communications in Applied Mathematics and Computational Science* 5.1 (2010), pp. 65–80. DOI: 10.2140/camcos.2010.5.65.

[3] Peter J. Green. "Reversible jump Markov chain Monte Carlo computation and Bayesian model determination". In: *Biometrika* 82.4 (1995), pp. 711–732. DOI: 10.1093/biomet/82.4.711.

[4] Alan Sokal. "Monte Carlo Methods in Statistical Mechanics: Foundations and New Algorithms". In: *Functional Integration: NATO ASI Series* 361 (1997), pp. 131–192.